THE ARTIHC STUDIO

WEBSITE DEVELOPMENT

PDS — Product Definition Sheet

Public Website & Admin Product Requirements

Version 2.0 | 01 October 2026

Prepared from the client brief, supplied portfolio, reference websites and the provided PDS/TSD/BAS examples.

1. Product Overview

A premium, modern-minimalist portfolio website for an interior/design studio. The website is a visual-first lead-generation product: visitors discover the studio, explore services and portfolio work, and submit a project enquiry.

2. Mandatory Pages

Page	Primary purpose	Core content
Home	Brand introduction + conversion	Hero, selected work, services, studio statement, process, CTA
About	Build trust	Studio story, philosophy, approach, locations/team if approved
Services	Explain offerings	Service cards, descriptions, supporting visuals, CTA
Portfolio	Showcase work	Project grid, category filters, project cards
Contact Us	Capture leads	Contact details, enquiry form, WhatsApp CTA, location/social links
Admin Login	Secure owner access	Email/username, password, authentication state
Admin Dashboard	Manage + monitor	Portfolio update, service update, GA4 and Microsoft Clarity monitoring links/summary

3. Homepage Requirements

- Full-bleed visual hero with concise positioning statement and one primary CTA.
- Featured projects selected from the admin panel.
- Services preview with clear links to Services.
- Short About/studio philosophy section.
- Editorial project storytelling using large images.
- Process section only if client confirms the actual process.
- Contact/Start a Project CTA repeated at logical points.
- Minimal footer with approved phone, email, locations and social links.

4. About Page

- Studio introduction and origin story using approved copy.
- Design philosophy and approach.
- Optional journey/timeline.
- Optional team information only if supplied by client.
- Locations and contact CTA.

5. Services Page

- Services are database-driven and editable from admin.
- Each service should have title, short description, detailed description, image/icon and display order.
- Service page should support a strong visual hierarchy rather than dense text.
- Every service section should lead naturally toward an enquiry.

6. Portfolio Page

The supplied 44-page portfolio is the primary content reference for the visual storytelling approach. It contains project images, project narratives, locations and project/type/scope information. For example, pages 4–21 show detailed residential projects and pages 22–43 continue with projects such as Montique, Farm Bijou, Cerulean Holt, The Office and The Azure. ■filecite■turn0file0■L9-L48■

Portfolio cards should feel editorial and image-led. Recommended fields: project title, category, location, type, scope, short description, cover image, gallery, featured flag and publication status.

7. Contact Us / Enquiry

- Name, phone, email, project type, location, approximate area, budget range (optional), message and consent.
- On successful submission, save the request to MySQL.
- Show a clear success state and WhatsApp action.
- WhatsApp message should contain the important enquiry fields in a readable format.
- Owner can later change enquiry status in the admin panel.

8. Admin Product Requirements

- Admin login page is separate from public navigation.
- Dashboard must provide quick access to Portfolio, Services and Analytics.
- Portfolio management: add/edit/delete, draft/publish, featured, gallery upload/reorder.
- Services management: add/edit/delete, publish/unpublish and reorder.
- Traffic monitoring: Google Analytics 4 and Microsoft Clarity.
- Enquiries: view stored client requests and status.
- Only authorized owner/admin accounts can access the panel.

9. Design Ownership — Antigravity

Antigravity is given full creative control over the website's visual design and implementation. The developer should analyze the client's portfolio, the two reference websites, the studio's actual project imagery and the requirements in this PDS before creating the UI. The design should not be a pixel-for-pixel copy of either reference site.

Design freedom includes layout, typography pairing, spacing system, card treatment, transitions, image composition, navigation behavior, responsive behavior and visual storytelling. The non-negotiables are usability, responsiveness, performance, accessibility, brand/content accuracy and the required pages/features.

10. Visual Direction

Target character: sophisticated, architectural, calm, premium, editorial, minimalist and image-led. Avoid generic corporate dashboards, excessive gradients, template-like cards, heavy shadows and visually noisy animations.

The supplied portfolio itself uses a restrained editorial presentation: white space, dark typography, warm wood/stone imagery, architectural compositions and large photographs. This should inform the mood without forcing the website to copy the PDF layout. ■filecite■turn0file0■L6-L18■

11. Acceptance Summary

- All seven required areas are implemented: Home, About, Services, Portfolio, Contact Us, Admin Login and Admin Dashboard.
- Portfolio and services are editable without changing source code.
- Enquiries are stored in MySQL.
- GA4 and Microsoft Clarity are integrated or securely linked according to the final implementation.
- WhatsApp enquiry handoff works as agreed.
- Responsive behavior is verified on desktop, tablet and mobile.

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WEBSITE DEVELOPMENT

TSD — Technical Specification Document

Frontend, Backend, Database, Analytics, Security & Design System

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Prepared from the client brief, supplied portfolio, reference websites and the provided PDS/TSD/BAS examples.

1. Recommended Technology Stack

Layer	Recommendation	Why
Frontend	React + Vite	Modern component architecture and fast production build.
Backend	PHP 8.3 + PDO	Fits GoDaddy cPanel shared hosting and MySQL.
Database	MySQL	Matches the client's requested database and hosting environment.
Admin UI	React-based admin	Consistent UI and reusable components.
Hosting	GoDaddy cPanel / Linux	Matches current client requirement.
Analytics	Google Analytics 4	Traffic, acquisition, engagement and conversion events.
Behavior analytics	Microsoft Clarity	Session recordings, heatmaps and behavior insights.
Source control	Git + GitHub	Version history and rollback.

2. Architecture

Recommended production architecture: Browser → React/Vite frontend → PHP API → MySQL. The public frontend can be served as static build files, while PHP handles authentication, CRUD, enquiry submission, analytics-related configuration and database operations. This avoids requiring a permanently running Node.js server on GoDaddy shared hosting.

3. Backend Architecture — BAS Boundary

The backend is divided into five logical modules: Authentication, Portfolio Management, Services Management, Enquiry Management and Analytics/Integration. File/image handling is treated as a supporting infrastructure module.

Module	Responsibilities
Authentication	Admin login, session management, logout, authorization checks.
Portfolio	Project CRUD, categories, publish status, featured projects, gallery metadata.
Services	Service CRUD, descriptions, media, order and publish status.
Enquiries	Validate, store and retrieve client requests; update lead status.
Analytics	Provide secure admin access/summary for GA4 and Microsoft Clarity; optional custom event/pageview summary.
Media	Upload validation, file naming, storage paths, image metadata and deletion rules.

4. Database Schema

Table	Key fields	Purpose
admins	id, email, password_hash, role, last_login	Admin authentication
projects	id, title, slug, category_id, type, location, area, scope, year, short_desc, full_desc, cover_image, featured, status	Portfolio projects
project_images	id, project_id, image_path, alt_text, sort_order	Project gallery
categories	id, name, slug, status, sort_order	Portfolio filtering
services	id, title, slug, short_desc, full_desc, media, sort_order, status	Services
enquiries	id, name, phone, email, project_type, location, area, budget, message, status, created_at	Client requests
settings	id, key, value	Site/brand configuration

5. API Design

Method	Endpoint	Access	Purpose
POST	/api/auth/login	Public	Authenticate admin
POST	/api/auth/logout	Admin	End session
GET	/api/projects	Public	Published projects

Method	Endpoint	Access	Purpose
GET	/api/projects/{slug}	Public	Project details
POST / PUT / DELETE	/api/projects...	Admin	Manage projects
GET	/api/services	Public	Published services
POST / PUT / DELETE	/api/services...	Admin	Manage services
POST	/api/enquiries	Public	Create enquiry
GET / PUT	/api/enquiries...	Admin	View/update enquiries
GET	/api/analytics	Admin	Optional internal summary

6. Admin Dashboard Technical Behavior

Dashboard should not attempt to recreate the entire GA4 or Clarity platforms. Instead, it should provide a clean executive overview and/or secure links/embedded summaries where technically appropriate.

Area	Implementation
GA4	Track page_view plus custom events such as project_view, enquiry_start, enquiry_submit and whatsapp_click.
Clarity	Install Clarity tracking script on approved public pages; use Clarity for recordings, heatmaps and behavior analysis.
Dashboard cards	New enquiries, published projects, published services and selected traffic metrics.
Analytics access	Owner-only. Never expose analytics credentials or API secrets in frontend code.

7. Classic + Modern Color Directions

Antigravity has full design control, so these are design directions rather than a rigid UI specification. The selected palette should be validated against the actual portfolio photography and logo.

Direction	Primary	Secondary	Accent	Background	Text
Classic Warm Luxury	#1F1D1A	#8C7355	#B28A5A	#F7F4EE	#24211E
Modern Architectural	#171717	#6B6B63	#A68B6B	#FAFAF8	#1E1E1E
Earthy Contemporary	#2B2A25	#7B7569	#A56B4F	#F3EFE8	#25231F
Monochrome Editorial	#111111	#555555	#A89F92	#FFFFFF	#161616

Recommended visual approach: start from the Modern Architectural or Classic Warm Luxury direction, then let Antigravity tune the exact tones after analyzing the logo and project photographs. Use one restrained accent rather than multiple competing accent colors.

8. Typography & UI Design Freedom

Antigravity should select a refined heading/body pairing suitable for an architecture/interior brand. Serif + clean sans, or two complementary sans families, are both acceptable. Avoid excessive font weights. Use a generous spacing scale, strong image ratios, restrained borders and subtle motion.

9. Security

- Use password_hash/password_verify for admin credentials.
- Use secure HTTP-only sessions and CSRF protection.
- Use prepared PDO statements.
- Validate all input server-side.
- Restrict uploads by MIME type, extension, size and dimensions.
- Keep DB credentials and analytics/API secrets outside frontend source.
- Protect admin routes with authorization middleware.
- Rate-limit enquiry submissions and add spam protection.

- Force HTTPS in production.
- Back up database and uploaded media.

10. Deployment

- Create MySQL database/user in GoDaddy cPanel.
- Verify PHP version and extensions.
- Build React/Vite production bundle.
- Deploy frontend and PHP API.
- Import database schema and create initial admin.
- Configure upload directories and permissions.
- Configure domain and HTTPS.
- Install GA4 and Clarity IDs.
- Test production forms, database writes, analytics events and WhatsApp flow.

11. Performance

- Use WebP/optimized JPEG images and responsive image sizes.
- Lazy-load non-critical portfolio images.
- Keep homepage media selective rather than loading the entire portfolio.
- Minify production assets and use caching.
- Limit animation and third-party scripts.
- Test Core Web Vitals after real portfolio images are uploaded.

12. Key Technical Decision

Do not use a Node/Express backend for the first version if the client's GoDaddy plan is standard shared cPanel hosting. React/Vite + PHP + MySQL is the lower-risk deployment architecture for the stated environment. A Node backend can be considered later if the client moves to hosting that supports a persistent Node runtime.

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BAS — Backend Architecture Specification

Server Architecture, Modules, Data Flow & Integrations

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Prepared from the client brief, supplied portfolio, reference websites and the provided PDS/TSD/BAS examples.

1. Architecture Overview

The backend is the secure application layer between the public React website/admin interface and MySQL. Its responsibilities are authentication, business validation, database operations, media handling and controlled integrations.

2. High-Level Flow

Public User → React Frontend → PHP REST API → MySQL. Admin User → Admin React UI → PHP REST API → MySQL. Analytics scripts run on approved pages and send measurement data to Google Analytics 4 and Microsoft Clarity.

3. Core Backend Modules

Module	Responsibilities	Main data
Authentication	Login, logout, session, authorization	admins
Portfolio	CRUD, publishing, featured status, gallery	projects, project_images, categories
Services	CRUD, publishing and ordering	services
Enquiries	Validation, database save, status changes	enquiries
Media	Upload validation, storage and metadata	project_images / project media
Analytics integration	GA4/Clarity configuration and optional summary	settings / analytics config

4. Authentication Architecture

- Single owner/admin account initially; support role field for future expansion.
- Login endpoint validates credentials and establishes a secure session.
- Passwords are hashed and never returned by API responses.
- Every admin CRUD request checks authentication and authorization.
- Logout destroys the session.
- Optional login throttling should be enabled to reduce brute-force attempts.

5. Portfolio Service

The portfolio service provides category-based project retrieval, project detail retrieval by slug, admin CRUD, featured-project selection and gallery management.

Operation	API	Database action
List public	GET /api/projects	SELECT published projects
Detail	GET /api/projects/{slug}	SELECT project + gallery
Create	POST /api/projects	INSERT project
Update	PUT /api/projects/{id}	UPDATE project
Delete/archive	DELETE /api/projects/{id}	Archive/delete with media cleanup rules
Gallery	POST/DELETE /api/projects/{id}/images	Manage project_images

6. Services Service

Services follow the same controlled CRUD pattern. Only published services are exposed to the public website. Admin controls title, descriptions, media, order and status.

7. Enquiry Service

Flow: validate request → spam/rate-limit checks → sanitize/normalize → INSERT into enquiries → return success response → generate WhatsApp action/message → owner follows up. The database record must be retained even if the WhatsApp action is not completed.

8. WhatsApp Integration

Phase 1: generate a WhatsApp click-to-chat URL with a prefilled message. Phase 2: if the client requires automated server-side delivery, integrate the approved WhatsApp Business/Cloud API separately. API credentials must remain server-side.

9. Analytics Architecture

Google Analytics 4 and Microsoft Clarity should be treated as external analytics systems rather than duplicating their complete functionality in MySQL. The website sends approved events to GA4 and includes the Clarity tracking script. The admin dashboard can show key business metrics and provide secure access to deeper analytics.

Suggested GA4 events: page_view, project_view, service_view, enquiry_start, enquiry_submit, whatsapp_click.

Suggested Clarity usage: session recordings, heatmaps and behavior observations to understand navigation and enquiry friction.

10. Database Relationships

Relationship	Rule
categories → projects	One category can contain many projects.
projects → project_images	One project can contain many images.
admins → admin actions	Future audit logging can associate actions with an admin.
enquiries	Independent lead records; status supports follow-up.

11. API Response Standard

Use consistent JSON responses such as { success, message, data, errors }. Never expose SQL errors, file-system paths, credentials or stack traces to public clients.

12. Media Architecture

- Upload original files to a controlled server directory.
- Validate MIME type, extension, size and image dimensions.
- Generate safe unique filenames.
- Store only the path/metadata in MySQL.
- Use optimized versions for public pages where possible.
- Prevent direct execution of uploaded scripts by server configuration.

13. Security Controls

- Prepared SQL statements.
- CSRF protection for authenticated state-changing requests.
- Secure cookies and session expiry.
- Authorization on every admin endpoint.
- Server-side validation regardless of frontend validation.
- Rate limiting/spam protection for enquiries.
- HTTPS only in production.
- Regular backups and controlled admin credential recovery.

14. Error Handling

Scenario	API behavior
Invalid login	Generic authentication error; no credential details.

Scenario	API behavior
Invalid form	400 validation response with field-level errors.
Unauthorized admin	401/403 response.
Missing record	404 response.
Database failure	500 response with generic message; detailed error logged server-side.
Upload failure	Validation/error response; no partial record left behind.

15. Backend Acceptance Criteria

- Admin login protects all admin APIs.
- Portfolio CRUD works and only published records appear publicly.
- Gallery records remain correctly linked to projects.
- Services CRUD works and published services appear publicly.
- Enquiries are reliably stored in MySQL.
- WhatsApp message generation contains approved enquiry fields.
- GA4 and Clarity tracking is present on intended public pages.
- No secret keys are exposed in frontend code.
- Production API returns controlled JSON responses and logs server-side errors.

16. Hosting Constraint

The backend specification intentionally avoids a continuously running application server because the client has specified GoDaddy web hosting. If the GoDaddy plan later supports or is upgraded to a suitable Node/Python runtime, the API layer can be migrated, but this is not required for version 1.